

Hosting Capacity Assessment of Distribution Networks with PV and EV Charging Stations

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Abstract—Explosive growth of distributed photovoltaic (PV) and electric vehicles (EV) imposes bidirectional power flows and voltage violations on distribution networks. Hosting capacity is redefined as the maximum aggregation of the capacity of PV and EV charging stations that can be accommodated. A capacity-maximization problem is formulated subject to four joint constraints: PV capacity, EV chargers' capacity, curtailment rate, and power flow limits. The non-convex Distflow model is relaxed via second-order cone programming to yield a tractable optimal power flow model. Reverse load rate and nodal voltage deviation are selected as evaluation metrics to access the hosting capacity. The case studies based on IEEE 33-node simulations return a hosting capacity of 35.90 MVA under the typical scenario, with reverse load rate below 80% and voltage deviation within $\pm 7\%$. The study provides operational guidance for secure network operation and renewable integration, while underscoring the need for a comprehensive assessment framework in future work.

Keywords—hosting capacity of distributed network, SOCP, Distflow, photovoltaic (PV), electric vehicle (EV)

I. INTRODUCTION

Distribution networks now face bidirectional power flows and temporal-spatial mismatch between distributed generation and demand. High-penetration photovoltaic (PV) has saturated hosting capacity in parts of China, while uncoordinated electric vehicle (EV) charging deepens peak-valley differences, frequently pushing feeders into overload and voltage violation. Quantifying the maximum hosting capacity is therefore essential for secure operation and integrating renewables in China's next-generation power system [1, 2].

In order to quantify the system's acceptance capacity of distributed resources, domestic and foreign scholars have proposed the concept of "hosting capacity (HC)", which initially focuses on PV access scenarios and evaluates the maximum

acceptance capacity of the system without violating voltage, thermal stability, and other constraints through deterministic power flow, stochastic simulation, or optimization algorithms [3, 4]. However, with the rapid increase in EV penetration, studies have gradually shifted from "single PV" to "PV+EV" joint scenarios. It is found that disorderly charging will significantly reduce the access space of PV, while coordinated control based on vehicle to grid (V2G) or orderly charging strategies can effectively mitigate voltage fluctuations and peak-to-valley differences, and improve the overall carrying capacity of the system [5, 6]. Furthermore, existing studies have pointed out that the traditional evaluation method based on voltage or ampacity as a single criterion is difficult to meet the needs of refined planning in the context of high proportion of new energy and high elastic load. It is urgent to build a bearing capacity evaluation system that integrates multi-dimensional indicators such as voltage quality, thermal stability, harmonic level, reliability and economy, and introduce active management methods such as energy storage configuration, network reconstruction, and reactive power optimization to achieve improvement of HC and collaborative optimization [7, 8]. The above research provides a theoretical basis and methodological support for the multi-dimensional carrying capacity assessment and improvement strategy of distribution network for PV and EV collaborative access.

Overall, although single-resource and multi-resource hosting capacities have been examined from selected technical angles, three gaps remain: 1) the constraint system of PV and EV charging stations combined hosting capacity needs to be further integrated, such as abandonment rate, civil capacity limit of charging piles; 2) the non-convex nature of the tide model requires an efficient linearization method to balance the calculation accuracy and efficiency; 3) the evaluation indicators need to be more in line with the actual operating standards, such as the national standard threshold for reverse load rate and

voltage deviation. In this context, this paper quantifies the combined hosting capacity of PV and EV charging by extending the Distflow model with second-order cone relaxation. Four linearized constraints, i.e. PV capacity siting, charger sizing, curtailment rate and power flow, are imposed and hosting limits are identified through reverse load rate and nodal voltage deviation.

II. ASSESSMENT MODEL OF DISTRIBUTION NETWORK HOSTING CAPACITY

The HC of distributed power generation refers to the maximum installed capacity of the power system that can be connected to distributed power generation under the condition of meeting the safety of power grid operation and power quality, taking into account factors such as load level, renewable energy utilization, and flexible adjustment of resources.

A. Objective Function

In accordance with the above industry rules, the time granularity is set to a coarse level, with hours as the unit of measurement. The hosting capacity of the distribution network for PV and EV charging stations is scientifically evaluated by analyzing the impact of the overlap between the power output of PV and the peak charging period for EV within the 24-hour period of a typical day. Then maximize the aggregated PV and EV chargers' capacities at all nodes while satisfying security constraints, that is:

$$\max \left(\sum_{i \in I^{PV}} S_i^{PV} + \sum_{i \in I^{EVCS}} N_i^{EVCS} S_i^{EVCS} \right) \quad (1)$$

where, S_i^{PV} is the hosting capacity of PV, and S_i^{EVCS} is the single EV charger capacity, and N_i^{EVCS} is the maximum number of EV chargers at the EV charging stations of node i .

B. Constraints

Hosting capacity assessment of distribution network must enforce nodal PV limits, EV charging station ratings and power-flow feasibility [9].

1) Capacity constraints of distributed PV

To prevent line and equipment from overloading the endanger secure operation, the nodal PV output $P_{i,t}^{PV}$ and its curtailment $P_{i,t}^{PV,cur}$ cannot exceed the local hosting limit $\bar{P}_{i,t}^{PV}$:

$$\begin{cases} P_{i,t}^{PV} + P_{i,t}^{PV,cur} = \bar{P}_{i,t}^{PV} \\ \bar{P}_{i,t}^{PV} \leq S_i^{PV} \\ P_{i,t}^{PV,cur} \geq 0 \end{cases} \quad (2)$$

Due to roof area, type and investment cost, the installed PV capacity S_i^{PV} at node i is capped by the maximum capacity $S_{i,max}^{PV}$ that the node i can host at any time:

$$S_i^{PV} \leq S_{i,max}^{PV} \quad (3)$$

The actual PV output $P_{i,t}^{PV}$ is governed by irradiance:

$$P_{i,t}^{PV} = \xi S_i^{PV} \frac{I_t}{I_0} \quad (4)$$

where I_t is the actual irradiance, and I_0 is the standard test condition irradiance, and ξ is the conversion efficiency.

During DC to AC conversion, the inverter supplies reactive power to control the power factor angle.

$$\left\| \begin{matrix} \bar{P}_{i,t}^{PV} \\ Q_{i,t}^{PV} \end{matrix} \right\|_2 \leq S_i^{PV} \quad (5)$$

2) Capacity constraints of EV chargers

The number of EV chargers deployed at node i is limited by the civil work and urban planning constraints:

$$0 \leq N_i^{EVCS} \leq \bar{N}_i^{EVCS} \quad (6)$$

The actual charging load $P_{i,t}^{EVCS}$ at node i cannot exceed the installed charger rating $N_i^{EVCS} S_i^{EVCS} \cos \varphi_i$, and the total charger capacity is further limited by the maximum load $P_{i,t,max}^{EVCS}$ that the node can host at any time:

$$P_{i,t}^{EVCS} \leq N_i^{EVCS} S_i^{EVCS} \cos \varphi_i \leq P_{i,t,max}^{EVCS} \quad (7)$$

Invoking the voltage-drop principle under the extreme case of unity power factor [10], the EV load is treated as purely active with $\cos \varphi_i = 1$ to yield the largest power and the lowest voltage.

3) Constraints of curtailment rate

The PV generation is intermittent during daylight hours. If local transfer capacity or demand is insufficient, surplus energy is curtailed [11]. To keep the hosting limit from relying on excessive curtailment, the curtailment rate is bounded:

$$\frac{P_{i,t}^{PV,cur}}{P_i^{PV} + P_{i,t}^{PV,cur}} \leq \bar{\xi}^{PV} \quad (8)$$

where, $\bar{\xi}^{PV}$ denotes the maximum allowable curtailment rate.

4) Constraints of power flow

The Distflow model was introduced for radial networks with high DER penetration. By linearizing the power flow equations, it reduces computational burden and rapidly quantifies impacts of distributed generation on voltage profiles, line flows and losses, thereby supporting minimization of voltage control loss and planning of distribution system.

A radial distribution network based on the Distflow formulation is illustrated below.

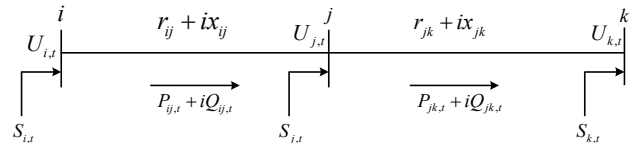


Fig. 1. Equivalent model of a radial distribution network.

In this figure, i , j , and k are the node number, $r_{ij} + ix_{ij}$ and $r_{jk} + ix_{jk}$ are the branch impedance.

The Distflow branch-based power flow equations are given below.

$$P_{ij,t} = r_{ij} l_{ij,t} + \sum_{k \in \Theta_j} P_{jk,t} + P_{j,t} \quad (9)$$

$$Q_{ij,t} = x_{ij} l_{ij,t} + \sum_{k \in \Theta_j} Q_{jk,t} + Q_{j,t} \quad (10)$$

$$v_{j,t} = v_{i,t} - 2(r_{ij} P_{ij,t} + x_{ij} Q_{ij,t}) + (r_{ij}^2 + x_{ij}^2) l_{ij,t} \quad (11)$$

$$\left\| \begin{matrix} 2P_{ij,t} \\ 2Q_{ij,t} \\ l_{ij,t} - v_{i,t} \end{matrix} \right\|_2 \leq l_{ij,t} + v_{i,t} \quad (12)$$

$$V_j^2 \leq v_{j,t} \leq \bar{V}_j^2 \quad (13)$$

$$0 \leq l_{ij,t} \leq \bar{l}_{ij}^2 \quad (14)$$

$$P_{j,t} = P_{j,t}^L - P_{j,t}^{PV} + P_{j,t}^{EVCS} \quad (15)$$

$$Q_{j,t} = Q_{j,t}^L - Q_{j,t}^{PV} \quad (16)$$

where, $P_{j,t}$ and $Q_{j,t}$ denote the active and reactive power injected at node j at time t ; $P_{ij,t}$ and $Q_{ij,t}$ are the corresponding branch flows at the sending end of line ij ; $v_{j,t}$ is the squared nodal voltage; $l_{ij,t}$ is the squared branch current.

III. ASSESSMENT INDEX OF DISTRIBUTION NETWORK HOSTING CAPACITY

This paper analyzes from the perspective of power supply reliability and power supply quality rate of distribution network, taking into account the growth of basic load, electric vehicle and photovoltaic penetration rate of distribution network, and selects two key indicators to evaluate the bearing capacity of distribution network.

A. Reverse load rate

The Reverse load rate κ_t of a distribution network is defined as the rate of reverse power flow through the transformer to its rated capacity. It can be expressed as:

$$\kappa_t = \frac{\sum_{i=1}^m P_{i,t}^{DG} - \sum_{j=1}^n P_{j,t}^{EV} - \sum_{l=1}^k P_{l,t}^{Load}}{S_N} \times 100\% \quad (17)$$

where, S_N denotes the rated capacity of the transformer at the slack bus. According to the national standard of China, $\kappa_t < 0$ indicates full local consumption without reverse flow and is recommended for new installations; $0 \leq \kappa_t \leq 80\%$ allows accessment provided the reverse flow remains below 80 % of the equipment rating; $\kappa_t > 80\%$ prohibits any further capacity accessment.

B. Nodal voltage deviation

Nodal voltage deviation is the percentage difference between the actual voltage at a node and the system nominal voltage under normal operating conditions.

$$\Delta U_{i,t} = \frac{U_{i,t} - U_N}{U_N} \times 100\% \quad (18)$$

where, $\Delta U_{i,t}$ denotes the voltage deviation at node i during period t . Under GB/T 12325-2008, three-phase supply at or below 20 kV must remain within - 7% and +7% of the nominal value.

IV. CASE STUDIES

The proposed model is tested on the IEEE 33-node distribution system with a 10 MVA base and 12.66 kV base voltage [12]. PV units are placed at nodes 4, 11, 16, 19, 22 and 26; EV chargers are installed at nodes 2, 8, 12 and 30. The distribution network system diagram is shown below.

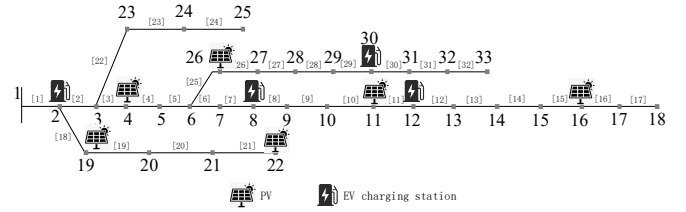


Fig. 2. The IEEE 33-node distribution network system diagram.

The maximum curtailment rate is set to 5 %. The installed capacity of a single EV charging station is 0.15MVA. The PV conversion efficiency is 0.18. Other parameters are provided in [13, 14].

A. Results analysis

Using a typical 24-hour period as a scheduling cycle, the study calculates the maximum amount of new energy that the distribution network can handle at the current time.

The hosting capacity evaluation yields an aggregate limit of 35.90MVA, comprising 13.10MW PV and 22.80MW EV chargers. Their bus-level capacities are displayed in Fig. 3 and Fig. 4.

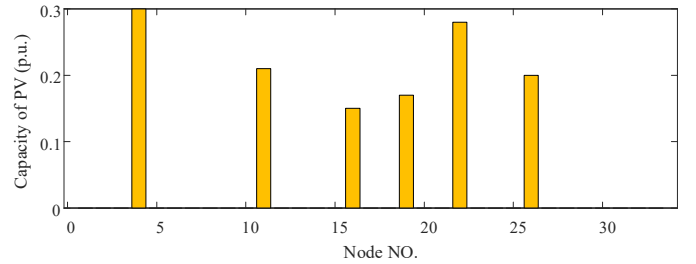


Fig. 3. The bus-level capacity of PV.

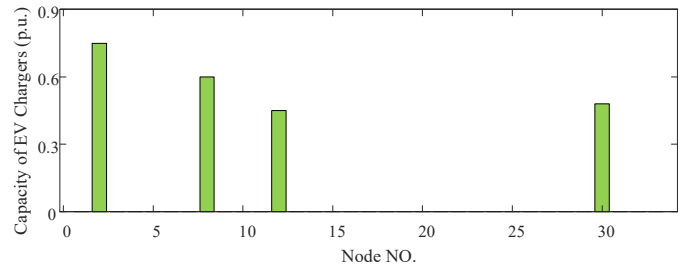


Fig. 4. The bus-level capacity of EV chargers.

Two conclusions emerge from the results in IEEE 33-node system.

First, buses on the main feeder and close to the substation host more PV because the low impedance and strong voltage support seldom violate limits; For example, node 4 accommodates 3MVA.

Second, end-of-feeder nodes face higher impedance and lower voltage, so added EV load readily triggers undervoltage or conductor overload, capping charger deployment. For example, node 12 is limited to 4.5MVA with 30 vehicles.

The time-varying active power profiles of PV for all nodes are shown in Fig. 5.

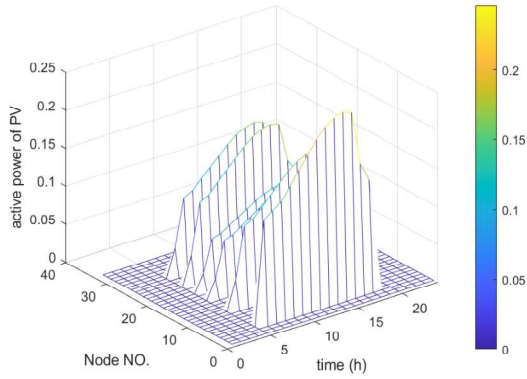


Fig. 5. The time-varying active power profiles of PV.

Under clear-sky conditions the PV output follows a single-peaked curve that tracks irradiance. Production climbs until 12–13h when the sun is highest, reaching its daily maximum. The curve remains smooth and stable, offering the grid a firm source of mid-day energy.

B. Indices analysis

1) Reverse load rate

Distributed PV integration converts the network from single-source to multi-source and alters both magnitude and direction of line flows. To illustrate the effect, Fig. 6 plots the active power flow along branches 2–17 (main feeder, slack-side branch excluded) at 10:00, 12:00, 14:00 and 20:00 on a clear-sky day.

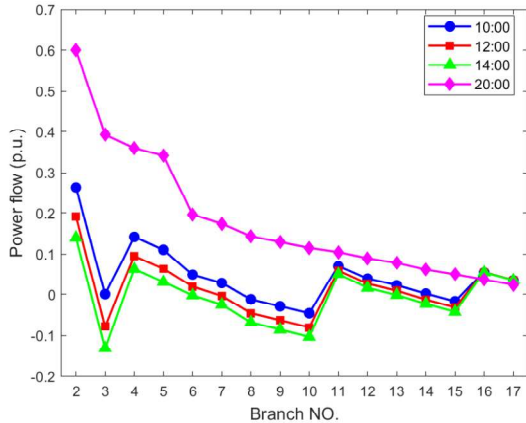


Fig. 6. The active power flow along branches 2–17.

With PV connected at buses 4, 11 and 16, negative flows appear on branches 3, 10 and 15 around midday. The reversed power is fully absorbed upstream because the PV rating is small enough to keep the net injection at each parent node positive and no back-feed into the substation occurs. At 20:00, without solar output, power decays monotonically from the source toward the loads and all branch flows remain positive.

At noon, the aggregated PV output equals 0.8970 p.u., EV charging 0.4889 p.u. and base load 0.3158p.u.; with a 5 MVA (0.5 p.u.) transformer the resulting reverse load rate is 18.46 %, well below the 80% threshold. Hour-by-hour computation for 1–24h confirms every value remains under 80%, validating the derived hosting capacity.

2) Nodal voltage deviation

With the integration of PV and EV charging stations, the voltage profile varies with time, location and size, as shown in Fig. 7. From 11:00 to 15:00, when PV output peaks, voltages rise and buses 4, 11, 19 and 26 approach the 1.07 p.u. limit. During the evening charging window (18:00–22:00) simultaneous EV demand depresses voltage, whereas the post-22:00 valley allows recovery. Since the terminal nodes are far from the upper-level substations, there is a significant voltage drop due to the resistance of the transmission lines, resulting in lower terminal node voltage. Throughout the day all nodal voltages remain within $\pm 7\%$ of nominal, confirming that the derived hosting capacity is acceptable under the voltage-deviation criterion.

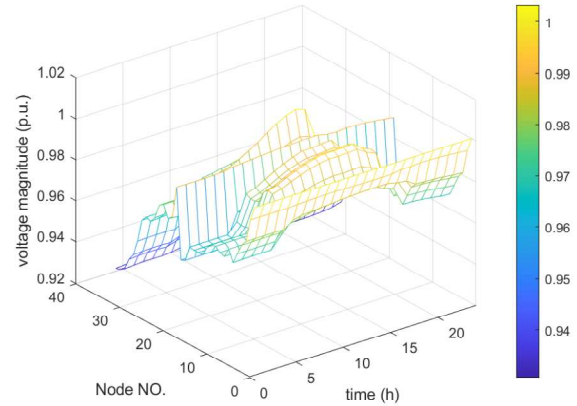


Fig. 7. Node voltage distribution diagram.

V. CONCLUSION

Large-scale PV and EV integration is addressed by maximizing installed capacity under locational, curtailment and operational constraints. The non-convex Distflow model is relaxed to a second-order cone program and the hosting capacity of the distribution network is computed. Main conclusions are as follows: 1) Simultaneous enforcement of PV siting capacity, EV charger, curtailment and power flow constraints captures all critical planning facets under renewable integration, yielding a practically relevant assessment model. 2) Reverse load rate and voltage compliance serve as quantitative proxies for network reliability and power quality, providing a representative and convincing measure of hosting capacity.

However, a systematic hosting-capacity framework that comprehensively validates the results and directly guides utilities in integrating PV and EV charging stations is still lacking and warrants future research.

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